

Alpha Diversity

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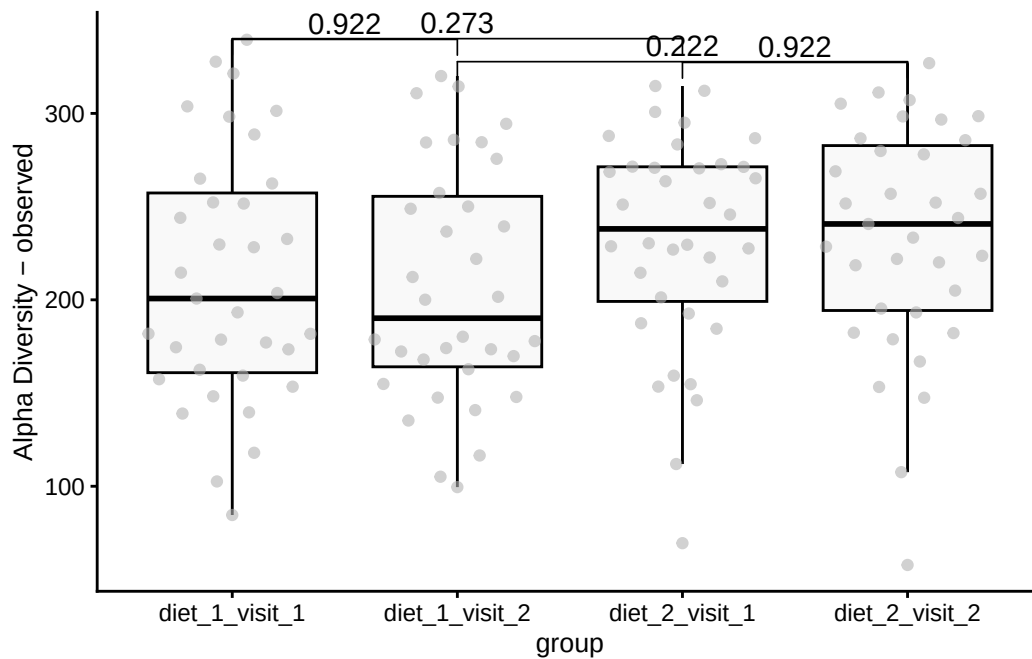
Introduction

This document reports on a microbiota project results of alpha diversity across different diets and visits.

Group-wise comparisons

- **Index:** observed
- **Group variable:** group
- **Multiple testing correction:** fdr

Visualization



Characteristic	after			before		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	144	59, 229	0.001	152	67, 236	<0.001
meal_group			0.3			0.4
1	—	—		—	—	
2	28	-16, 72	0.2	32	-11, 75	0.14
3	30	-14, 73	0.2	27	-15, 70	0.2
4	-3.8	-47, 39	0.9	10	-32, 53	0.6
age	1.2	-0.25, 2.7	0.10	1.0	-0.45, 2.5	0.2

¹CI = Confidence Interval

Paired test of before and after	Beta	95% CI ¹	p-value
(Intercept)	147	89, 206	<0.001
meal_group			0.11
1	—	—	
2	28	-2.7, 58	0.074
3	28	-2.5, 58	0.072
4	2.2	-28, 32	0.9
age	1.2	0.13, 2.2	0.026

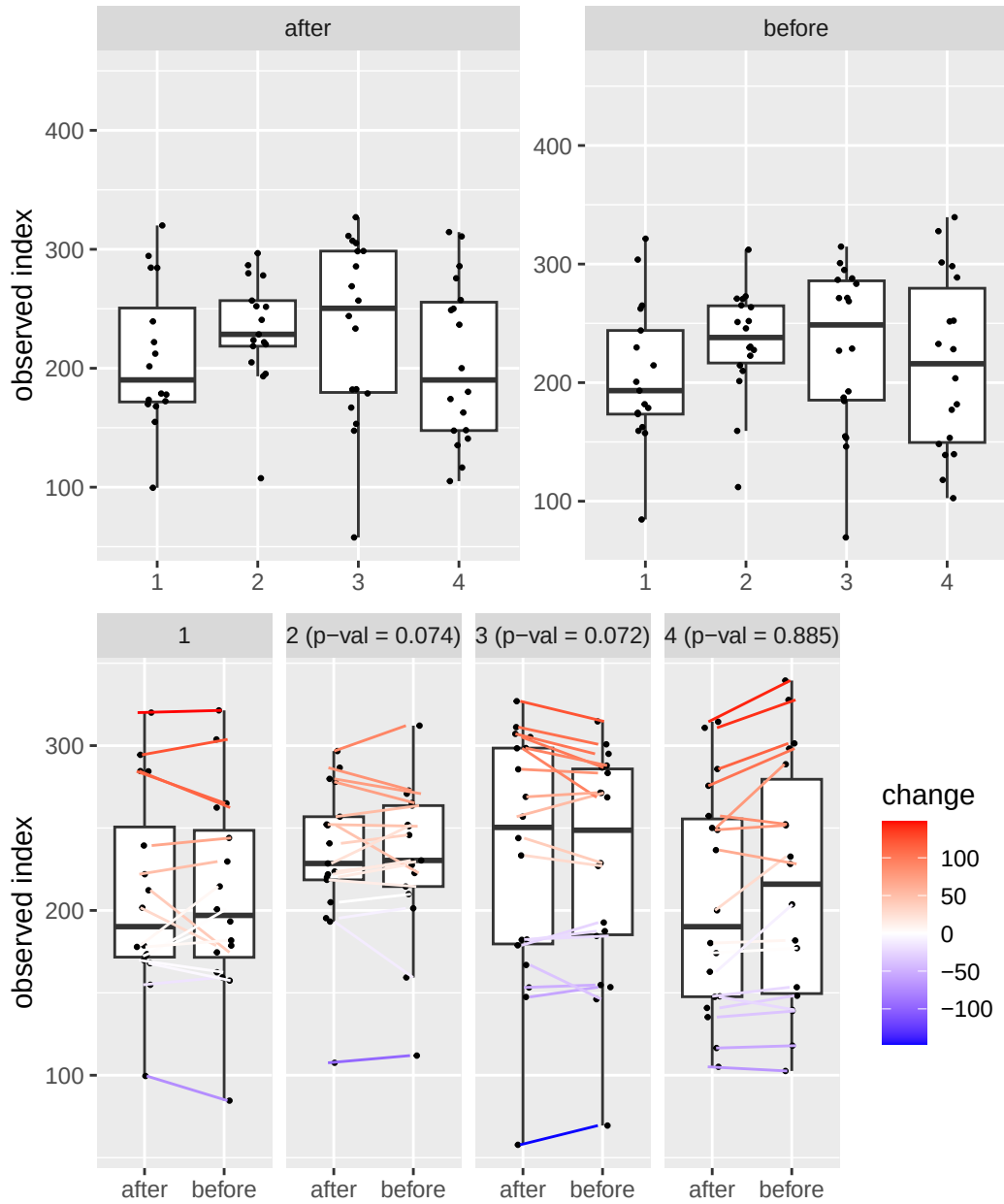
¹CI = Confidence Interval

Table of the top findings

Table 1: Table of alpha diversity comparisons

group1	group2	mean1	mean2	logFC	p.adj
diet_1_visit_1	diet_1_visit_2	211.16	207.15	-0.03	0.922
diet_2_visit_1	diet_2_visit_2	231.52	233.18	0.01	0.922
diet_1_visit_1	diet_2_visit_1	211.16	231.52	0.13	0.273
diet_1_visit_2	diet_2_visit_2	207.15	233.18	0.17	0.222

Alternate analyses



Data Summary

class: TreeSummarizedExperiment
dim: 1700 140

```
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```

Version info

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.1 LTS
```

```
Matrix products: default
BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0
```

```
locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

```
time zone: Europe/Helsinki
tzcode source: system (glibc)
```

```
attached base packages:
[1] stats4      stats      graphics  grDevices  utils      datasets  methods
[8] base
```

```
other attached packages:
 [1] kableExtra_1.4.0          vegan_2.6-8
 [3] lattice_0.22-6            permute_0.9-7
```

[5] lubridate_1.9.3	forcats_1.0.0
[7] stringr_1.5.1	purrr_1.0.2
[9] readr_2.1.5	tibble_3.2.1
[11] tidyverse_2.0.0	tidyr_1.3.1
[13] scatter_1.33.4	scuttle_1.15.4
[15] quarto_1.4.4	patchwork_1.3.0
[17] parameters_0.24.0	multtest_2.61.0
[19] miaTime_0.1.21	miaViz_1.15.4
[21] ggraph_2.2.1	mia_1.15.6
[23] TreeSummarizedExperiment_2.13.0	Biostrings_2.73.2
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[31] GenomeInfoDb_1.41.2	IRanges_2.39.2
[33] S4Vectors_0.43.2	BiocGenerics_0.51.3
[35] MatrixGenerics_1.17.0	matrixStats_1.4.1
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[39] ggpubr_0.6.0	ggplot2_3.5.1
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[43] cardx_0.2.2	ANCOMBC_2.8.0

loaded via a namespace (and not attached):

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